Calculation Drills - Answers

Full working shown. Marks in a real exam are given for method as well as the final answer, so always show your steps.

A. Added value

A1. 5 - 2 = $3 per loaf.

A2. 450 - 120 = $330 per table; total = 330 x 30 = $9,900.

A3. 9.99 - 1.20 = $8.79 per case.

B. Labour turnover

B1. (24/160) x 100 = 15%.

B2. (36/480) x 100 = 7.5%.

B3. 15% of 200 = 30 staff left.

C. Labour productivity

C1. 18,000/45 = 400 units per worker.

C2. 3,600/12 = 300 units per worker per week.

C3. 250 x 60 = 15,000 units.

D. Market share

D1. (6/48) x 100 = 12.5%.

D2. (90,000/500,000) x 100 = 18%.

D3. 12% of $250m = $30m.

E. Market growth

E1. ((92-80)/80) x 100 = 15%.

E2. ((360,000-400,000)/400,000) x 100 = -10% (a 10% fall).

E3. 150m x 1.08 = $162m.

F. Capacity utilisation

F1. (3,750/5,000) x 100 = 75%.

F2. (175/250) x 100 = 70%.

F3. 90% of 20,000 = 18,000 units.

G. Working capital

G1. 90,000 - 62,000 = $28,000 (positive - can meet short-term debts).

G2. 45,000 - 51,000 = -$6,000 (negative - a liquidity problem; may struggle to pay bills).

H. Total & average cost

H1. Total variable cost = 4 x 6,000 = $24,000. Total cost = 30,000 + 24,000 = $54,000. Average cost = 54,000/6,000 = $9 per unit.

H2. Total cost = 18,000 + (7 x 3,000) = 18,000 + 21,000 = $39,000. Average cost = 39,000/3,000 = $13 per unit.

I. Cash flow

I1. Net cash flow = 30,000 - 34,000 = -$4,000. Closing = 8,000 + (-4,000) = $4,000.

I2. Jan: net = 18,000 - 15,000 = +3,000; closing = 2,000 + 3,000 = $5,000. Feb: opening = $5,000; net = 16,000 - 20,000 = -4,000; closing = 5,000 - 4,000 = $1,000.

J. Contribution & profit

J1. 25 - 10 = $15 per unit.

J2. Total contribution = 15 x 4,000 = $60,000. Profit = 60,000 - 40,000 = $20,000.

J3. Contribution = 12 - 7 = $5. Total = 5 x 7,000 = $35,000. Profit = 35,000 - 25,000 = $10,000.

K. Break-even & margin of safety

K1. Contribution = 30 - 18 = $12. Break-even = 60,000/12 = 5,000 units.

K2. Margin of safety = 7,000 - 5,000 = 2,000 units. Profit = (12 x 7,000) - 60,000 = 84,000 - 60,000 = $24,000.

K3. Contribution = 20 - 11 = $9. Units for target profit = (fixed costs + target profit)/contribution = (45,000 + 18,000)/9 = 63,000/9 = 7,000 units.

L. Variances

L1. 86,000 - 80,000 = +$6,000 FAVOURABLE (revenue above budget).

L2. 57,000 - 50,000 = $7,000 higher costs = ADVERSE.

L3. 26,000 - 30,000 = -$4,000 ADVERSE (profit below budget).